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is also an instrumental step. Keeping in mind the energy requirements of sectors like ammonia, steel and cement, the NGHM lays a road map for providing green hydrogen at an economical cost for industrial decarbonisation beyond 2032.

Another important factor is energy storage, which is a critical component of reliable energy supply and grid stability. To address the challenge of intermittency of renewable power, the government has ensured that the majority of recent renewable—solar and wind—tenders have a mandatory inbuilt battery storage capacity target.

According to the Central Electricity Authority (CEA), India needs 336 GWh of storage by 2030, to be met largely by battery systems (208.25 GWh), with the rest being served by pumped hydro storage projects. As of September 2024, 211 MWh of BESS, or battery energy storage system, capacity is operational in India; 1.8 GWh of BESS is to be added in 2025.

INDIA'S ENERGY DEMAND

India's energy demand is seeing an unprecedented surge. Over the last three years, demand has grown at an accelerated rate, far exceeding historical averages.

Nomura has projected robust growth for India's power sector, anticipating a compound annual growth rate (CAGR) of over 7 per cent in electricity demand from FY24 to FY27. These mark a significant uptick compared to the moderate growth rate of 4–6% in the pre-pandemic years.

Pranav Master, Senior Practice Leader and Director, Consulting—Energy & Sustainability at independent research firm Crisil Intelligence, says RE capacity (excluding large hydropower) has grown at 13% CAGR between 2019